SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	#slaynergy
Project / Product Name	Jobbery
Final Live Deployment URL	http://sdc-runway.vercel.app/
Codebase Repository URL	https://github.com/yabgzichien/SDCHack-slaynergy (public, Apache-2.0 licence)
Name (Leader)	Yang Zi Chien
Email & phone (Leader)	zichienyang@gmail.com / 0139681431

2. Project Description

[Maximum 250 words]

Jobbery gives Malaysian fresh graduates and working adults a deadline-based career plan for the AI era. In June 2026 the Human Resources Minister said 697,000 jobs are highly affected by AI, digitalisation and the green economy unless workers upskill within three to five years, yet most tools give one fixed “AI risk” score from old research.

A user answers a short questionnaire or uploads a résumé. Jobbery then forecasts when AI could take over each of 80 skills, using dated public data from Epoch AI, the Anthropic Economic Index, the ILO and DOSM. It keeps three possible AI speeds (slow, medium and fast) and weights them by how well each predicted real past data releases, together with a points-only prediction market that settles on real releases. It ranks the jobs the user can reach before their deadline, with a six-part score the user can open and a list of what would raise it. It then builds a week-by-week plan from 175 checked free or low-cost courses, ending with a meeting with an advisor. Every role, skill and plan step is tagged to a UN SDG target, and the plan’s hours are totalled by target.

Generative AI only reads résumés and course certificates and helps find a moved DOSM release page. Code checks everything it proposes, and it never supplies a number. The server refreshes its data automatically, and a history picker replays four real DOSM releases. The prototype is live on Vercel with a Neon database.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Jobbery is one Next.js 16 and React 19 application written in TypeScript. It is deployed on Vercel (Singapore) with a Neon PostgreSQL database. Figure 1 shows how the parts connect.

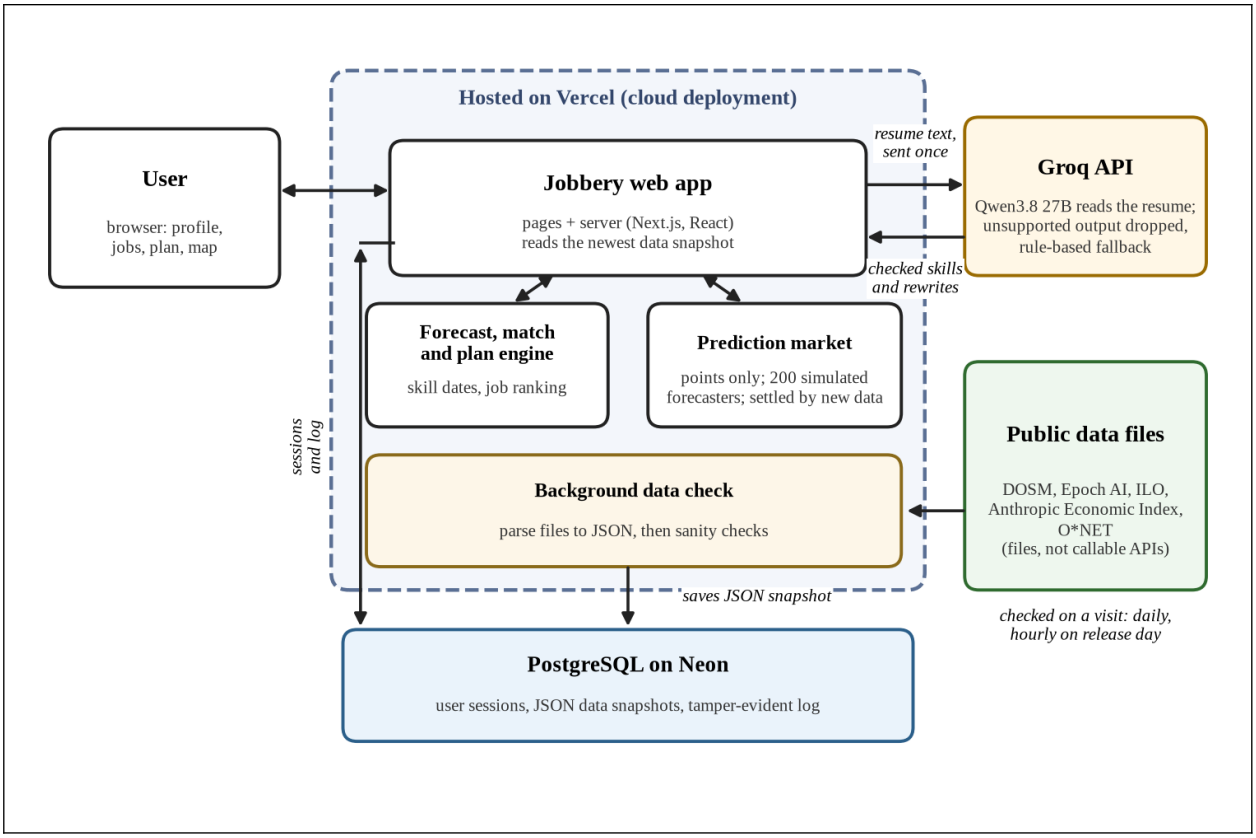
Front end. The screens (intake, skills radar, jobs, job map, plan, forecast and trust pages) are React pages styled with Tailwind CSS, with Recharts charts and a hand-drawn SVG map of Malaysia. The app is available in English, Bahasa Malaysia, Chinese and Tamil.

Engine. A plain, tested TypeScript module with no model in the loop holds the LMSR prediction market, about 200 simulated forecasters, the three AI futures and their weights, skill expiry dates, job matching and the plan builder. The same inputs always give the same answer. 249 automated tests check it, including a fairness test showing that age and gender never change the matches.

Data layer. DOSM, Epoch AI, the ILO, the Anthropic Economic Index and O*NET publish files, not APIs. A daily Vercel Cron job, plus visitor checks (hourly on a DOSM release day), downloads them, checks them against the previous numbers and saves them in Neon as numbered, hashed snapshots. If a check fails, the last good snapshot stays in use.

GenAI integration. When the user leaves the AI option on, the résumé text (or a course certificate) is sent once to a chain of providers: the KrackedDev AI API (DeepSeek V4.1 Flash) first, then Qwen2.5 14B on RunPod Serverless, then Qwen3.8 27B on the Groq API, and a rule-based scanner if all three fail. The model only proposes. The server drops any skill with no quoted résumé line, rejects any rewrite that adds a number and blocks injected instructions. The same chain helps find a moved DOSM release page and never supplies a number. The résumé and certificates are read in memory and never stored.

Database. Neon PostgreSQL (SQLite for local development) stores each visitor’s session, plan progress, advisor bookings, the data snapshots and a hash-chained activity log whose fingerprint is committed to the repository.



3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

What we built and deployed. All six journey stages run live from one URL (sdc-runway.vercel.app): profile intake by questionnaire or résumé, a skills radar with source and date on every number, ranked job matches with an openable score, a week-by-week plan with 175 hand-checked courses from 49 providers, SDG mapping with plan hours totalled by target, and a comparison of the three AI futures. Also built: a points-only prediction market, advisor booking, a job map with pay left after living costs, an illustrative employer page, a trust page and four languages. The code is public under Apache-2.0 with 249 passing tests.

What we changed from the Phase 1 design, and why. (1) The three AI speeds only changed skill dates, so we added an AI speed picker that re-ranks the jobs. (2) Data was static files, so the server now checks DOSM, Epoch AI and OpenDOSM itself, keeps the last good data if a check fails, and a history picker replays four real DOSM releases. (3) The job score gained a qualifications and experience part, and each job page shows “what would raise it”. (4) We replaced the fixed list of SDGs with tags on every role and skill, and now add up the plan’s hours by target. (5) A course step now finishes only with a certificate that the AI reads and plain code checks. (6) Groq’s free tier allowed about two résumé scans a minute, so KrackedDev and RunPod now come first. (7) One person moves a market answer by a fixed 0.1%. (8) A week-old session took 13 seconds to load and now takes about 0.4 seconds. (9) Vercel’s free plan rejects hourly cron jobs, so we use a daily cron plus visitor checks.

What we did not do: we have not run a user study.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Phase 1 baseline: 697,000 jobs highly affected by AI if workers do not upskill within three to five years. Phase 1 targets were 35,000 users, 90% still relying on the resources, 50% completing courses and quizzes, and 40% finding work within 3 years. These need real users over several years. We have not run a user study or built user counters, so we do not claim them. What the finished prototype measurably does, for the synthetic persona Hafiz (27, bank operations executive, 5 months, 10 hours a week): it ranks 59 roles and picks Procurement officer at a score of 72 out of 100. It builds a 10-week plan inside his 19-week deadline, with 79 plan hours, of which 56 are learning hours. All 25.5 course hours (100%) are free. The hours are totalled as 40 toward SDG 8.2, 39 toward 8.5, and 56 learning hours toward 4.4. The prediction market, replayed on real past releases, scores a Brier of 0.242 against 0.25 for a coin flip. The course list holds 175 courses from 49 providers, at least two for every skill.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs,*	The figures above were computed by running the app's own engine on the Hafiz persona (the same code the live site runs). 249 automated tests pass (2 skipped), run with npm test in the repository.

Metric Category	Your Response
hardware power-meter simulation readings, or database transaction counters.	Tests recompute the numbers by hand: the six score parts add up exactly to the score shown, each "what would raise it" gain equals the real change when that skill is added, the plan's SDG hours add up to the plan's total, and the market replay is deterministic. A fairness test shows that age, gender and home state never change the matches. The live Trust page (/trust) repeats the fairness check and verifies the hash-chained activity log. scripts/check-links.mjs re-checks every course link. Every data file carries its source, date and a hash.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Jobbery is free to the person, uses no accounts and stores no résumé or certificate. A person gets from "who I am today" to a sourced, SDG-tagged plan in minutes, using free courses (Hafiz's course hours are all free) with HRD Corp and PERKESO funding shown where it applies. The advisor step (for example an employment officer) fits into existing careers services. Likely partners are university and TVET career centres, HRD Corp and PERKESO, and employers' HR and learning teams, who are named in the problem. The employer page is illustrative demo data, not real staff data. No automated decision is made about anyone, and the person always chooses.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Running cost is near zero today: Vercel and Neon free plans, open public data (CC BY 4.0, ODbL or Malaysian Government Open Data Terms), and AI calls only when the user opts in, with a rule-based fallback if every provider is down. The engine is deterministic and cached, so a week-old session computes in about 0.4 seconds. It can be offered to institutions as a hosted service or as a skills-gap module added to a career-centre site. Limits: free-plan rate limits, simulated forecasters instead of real ones, and no user study yet.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

1. Job match score (src/lib/engine/matching.ts): score = Σ (part value × weight), with weights

```
WEIGHTS = { overlap: 0.25, outlook: 0.28, demand: 0.18, reach: 0.15, interest: 0.04, quality: 0.1 };
```

Hafiz's top job, Procurement officer, as computed by the engine:

```
0.25×0.500 + 0.28×1.000 + 0.18×0.673 + 0.15×0.684 + 0.04×0.500 + 0.10×0.717 = 0.7204, shown as 72/100 (no adjustment applied).
```

2. Prediction market price (src/lib/engine/lmsr.ts), Hanson's logarithmic market scoring rule with b = 60, points only:

```
cost(m) = b·ln(e^(qy/b) + e^(qn/b));   price(m) = 1 / (1 + e^((qn − qy)/b))
```

The most the market maker can lose on one question is b·ln 2 = 60 × 0.693 = 41.6 points. A person moves an answer by a fixed 0.1% however many points they stake, so one person cannot swing a market.

3. Forecast quality: Brier score BS = mean of (forecast − outcome)². In the replay on real past releases BS = 0.242, against 0.25 for always saying 50%.

4. SDG hours (src/lib/engine/plan.ts): a step that teaches k skills adds hours ÷ k to each skill's SDG target. For Hafiz: 40 h to 8.2 + 39 h to 8.5 = 79 plan hours, and 56 learning hours to 4.4 are counted separately. Tests check that the parts add up to the total.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Jobbery is already live, free to use and cheap to run, so the next step is data and trust, not rebuilding. With funding or partnerships (for example through SDC's Rotary network or a joint research project at XMUM), we would do five things.

First, run a pilot with university and TVET career centres, and with HRD Corp and PERKESO, to measure the Phase 1 KPIs on real users.

Second, scale the platform. The engine is deterministic and cached, so volume mainly needs paid hosting: Vercel Pro (which allows hourly data checks), a larger Neon database and a dedicated AI endpoint in place of free tiers. Data refresh is already automatic.

Third, replace the simulated forecasters with real ones: advisors, employers and graduates, still points-only, never money.

Fourth, build the employer view on real, consented staff data, and extend the task cards from finance to the other nine sectors.

Fifth, do an accessibility pass and add a low-bandwidth mode.

Revenue would come from institutions and employers, not from users. Individuals stay free, we do not sell data.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

Yes ☑️ No No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑️ Yes No No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16.4.2; React 19.3.0	Code	MIT
Tailwind CSS 4.3.3; Framer Motion 14.4.0; Recharts 3.10.1	Code	MIT
Lucide React 1.60.0	Code	ISC
TypeScript 5.9.3	Code	Apache-2.0
Node.js 22	Code	MIT
pg 8.23.1; better-sqlite3 13.0.3; fflate 0.8.3	Code	MIT
unpdf 1.9.1 (reads PDF resumes)	Code	MIT
mammoth 1.13.0 (reads Word resumes)	Code	BSD-2-Clause
Instrument Sans (font, via	Code (font)	SIL Open Font Licence 1.1

Google Fonts)		
DeepSeek V4.1 Flash (resume reader and data-source finder, via the KrackedDev AI API)	Model	The terms of the KrackedDev AI API and of the model's publisher apply
Qwen2.5 14B Instruct (second-choice resume reader, on RunPod Serverless)	Model	Apache 2.0 (open weights)
Qwen3.8 27B (last backup resume reader, via the Groq API)	Model	Apache 2.0 (open weights)
DOSM Job Market Insights, 2026 Q2	Dataset	Malaysian Government Open Data Terms of Use 1.0
OpenDOSM employment, unemployment and wage data by state	Dataset	CC BY 4.0
Epoch AI benchmarking data	Dataset	CC BY 4.0
Anthropic Economic Index (2025-02-10 release)	Dataset	CC BY (data)
ILO generative-AI exposure scores (2025 update)	Dataset	CC BY 4.0
O*NET occupation and task data	Dataset	CC BY 4.0
geoBoundaries Malaysia state borders (OpenStreetMap)	Dataset	ODbL 1.0
UN SDG target text (sdgs.un.org)	Dataset (text)	Quoted with source

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

Yes ☑️ No No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



Yang Zi Chien

31 Oct 2026